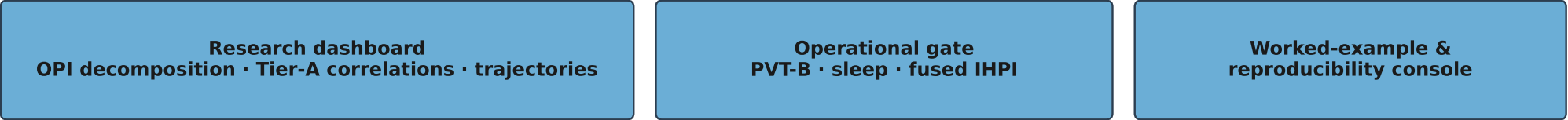
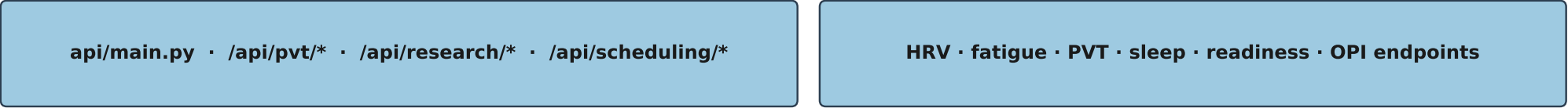


Figure 4 — Reference-implementation architecture

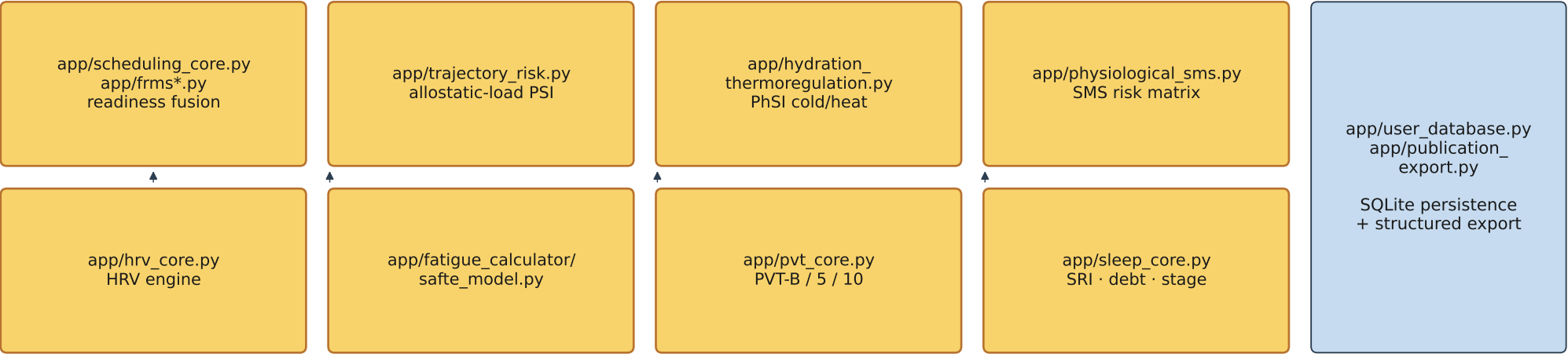
L4 — Next.js



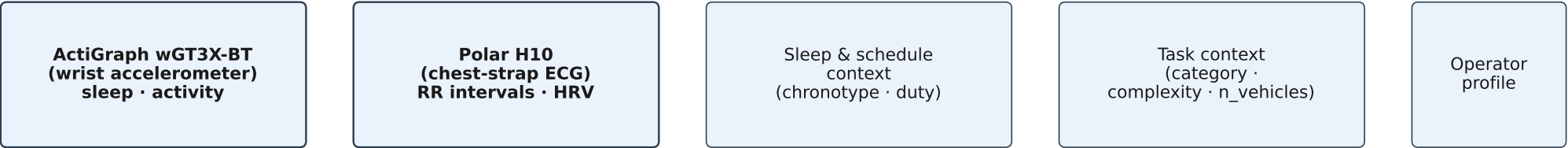
L3 — FastAPI



L2 — Python core



L1 — Inputs



Gold-bordered modules compose the OPI fusion pathway. Antarctic validation instruments boxed in bold at L1.